MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 5BCP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Miniaturized, magnetic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0	11,000	+180	1.5	80,000
Minimum:	5.7	-200	5,000	---	---	---
TEST CONDITIONS:	6.3	Adjust	6,000	---	---	---

GENERAL:

Qualification - Required (see note 1)

Ⓔ denotes changes

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1 To waive any portion of the referenced documents, or to amend contractual requirements

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5111	Vibration	-		Width	---	2.0	mm
Ⓔ 1141	Pressure implosion	-		---	---	---	---
Ⓔ 1002	Barometric pressure, reduced	2	20 mmHg	---	---	---	---
Ⓔ 1331	Direct-interelectrode capacitance	-	k to all g1 to all	Ck Cg1	--- ---	5.5 5.5	pF pF
Ⓔ 5216	Cathode illumination	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (magnetic types)	-		---	---	---	---
5101	Neck straightness	-	Cylinder length = 4 inches (101.6 mm); ID = 0.901 inches (22.89 mm) maximum	---	---	---	---
5106	Screen and faceplate blemishes	-		---	---	---	---
5223	Modulation	-	Ib = 200 µAdc	ΔEc1	---	34	Vdc
5221	Screens	-	Eb1 = 5,000 Vdc; Ib1 = 48 µAdc	cB1	640	---	cB
5231	Spot position (magnetic deflection)	-		---	---	6	mm
5236	Zero-bias anode current (magnetic deflection)	3		---	---	---	---
---	Focusing coil current	4	Ib = 200 µAdc D = 2-3/8 inches (60.33 mm)	---	48	72	mAdc
5241	Grid - cutoff voltage	-		---	-19	-56	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	270	330	mA
5101	Side terminal and base alinement	-	Halfway between pins No. 1 and No. 9	---	---	---	---
5101	Neck and bulb alinement (magnetic types)	5		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5101	Face tilt	5		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb1 = 11,000 Vdc	---	---	---	---
5221	Screen (P7 type)	-	Eb1 = 5,000 Vdc; Ib1 = 48 μAdc	---	---	---	---
5226	Line width A (magnetic deflection)	6	Ib = 200 μAdc	Width	---	0.33	mm
5251	Heater cathode leakage current	-	Ehk = +180 Vdc Ehk = -180 Vdc	Ihk Ihk	--- ---	15 15	μAdc μAdc
---	Astigmatism	7		Ratio	---	1.20	---
(E) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb1 = 11,000 Vdc; Ib = 60 μAdc; t = 500 hours	---	---	---	---
---	Life-test end points:						
5226	Line width A		200 μAdc	Width	---	0.33	mm
5223	Modulation	-	200 UAdc	ΔEc1	---	34	Vdc

NOTES:

1. The following information and materiel shall be made available to the government representative conducting the qualification inspection measurements with four regular qualification samples:

   (a) The gun drawing with significant dimensions.

   (b) A sample of the gun to be used in manufacturing of the tubes.

2. Maximum rated voltages are applied to all electrodes connected through the base, and with tube in cutoff conditions.

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NOTES: -Continued

3. The following values are to be used in place of those specified in method 5236:

Negative grid cutoff voltage	Minimum 0-bias current μA	Negative grid cutoff voltage	Minimum 0-bias current μA
19	250	38	480
20	250	39	500
21	250	40	520
22	250	41	540
23	250	42	560
24	250	43	580
25	250	44	600
26	270	45	620
27	285	46	640
28	300	47	660
29	310	48	680
30	335	49	700
31	350	50	725
32	370	51	750
33	390	52	775
34	405	53	800
35	425	54	825
36	445	55	850
37	460	56	875

4. Use JEDEC No. 127 focus coil, or equal. Focusing coil current shall be determined at the focus setting used in the astigmatism test.

5. The two sets of neck supports, referenced in method 5101, shall be spaced 1.125 inches (28.58 mm) and 3.375 inches (85.73 mm) below reference plane.

6. Line width A shall be the maximum line width obtained in the astigmatism test.

7. With a single line, 4.5 inches (114.30 mm) in length and 1/60-second duration, repeated at a rate of 60 Hz, modulate grid No. 1 with 1-microsecond pulses. Adjust the peak beam current to 200 μA and focus for the most circular spot at the center of the screen. Display a partially merged nonlinear raster and note the orientations for greatest and least mergings as the raster is rotated. Maintain this focus and proceed according to method 5226, to determine the line widths for those two orientations. Astigmatism (the ratio of the maximum to the minimum line width) shall not exceed the value specified.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 70, 80
DSA - ES

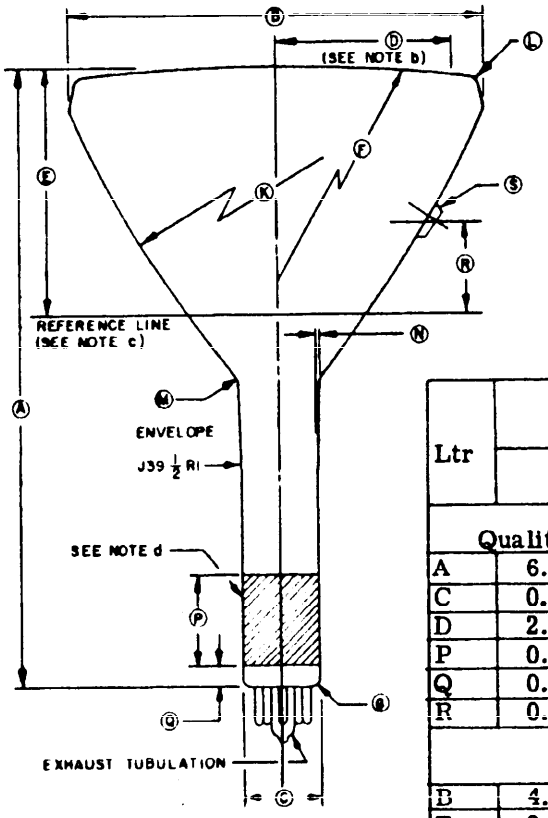
User activities:
Army - ME, WC, AV
Navy - AS, OS, CG, SH
Air Force - 11

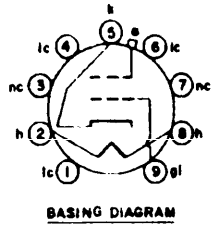
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3059)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.750 (171.45)	7.250 (184.15)
C	0.810 (20.57) Dia	0.896 (22.76) Dia
D	2.125 (53.98) R	
P	0.938 (23.83)	1.062 (26.97)
Q	0.188 (4.78)	0.250 (6.35)
R	0.563 (14.30)	0.813 (20.65)
Qualification inspection		
B	4.900 (124.46) Dia	5.000 (127.00) Dia
E	2.750 (69.85)	2.875 (73.03)
G	Base: E9-37 (EIA)	
S	Bulb contact: J1-22 (EIA)	
Reference dimensions (see note a)		
F	24.000 (609.60) R	
K	9.660 (245.36) R	
L	0.125 (3.18) R	
M	0.125 (3.18) R	
N	2°	

NOTES:
a. These dimensions are for information only and are not required for inspection purposes.
b. Useful screen.
c. Determined by point where leading edge of JEDEC reference line gage G123 will stop.
d. External conductive coating shall be tied to cathode potential.

FIGURE 1. Outline drawing of electron tube type 5BCP7.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVY DEPARTMENT
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NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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